ADITYA VIKRAM SINGH

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SUMMARY

Computer Science graduate with strong SQL, Python, data analysis, and AI engineering skills, experienced in building data pipelines, LLM-based workflows, and computer-vision solutions. Focused on transforming raw and unstructured data into reliable, business-focused insights and practical AI/data products.

EDUCATION

SRM Institute of Science and Technology **Jun 2022 - May 2026**
B.Tech, Computer Science and Engineering
- **GPA:** 9.38
- **Achievements:** Graduated First Class with Distinction

DPS R.K.Puram **2020 - 2022**
Class XII, PCM *New Delhi*
- **GPA:** 95.20%

PROJECTS

Recession Forecasting using Macroeconomic Indicators (FRED Time-Series Analysis) **Feb 2025 - May 2025**
- Built a quarterly recession forecasting dataset from FRED macroeconomic indicators including yield spreads, unemployment, CPI, and VIX, with structured train/validation/test splits for time-series modeling.
- Improved the baseline logistic regression approach using calibrated gradient boosting, producing clearer recession-risk signals for financial planning and risk analysis.
- Documented model assumptions, class imbalance, NBER chronology, and key drivers using a model card and permutation importance.

Financial Document Assistant with Retrieval-Augmented Generation (RAG on AWS) **May 2025 - Aug 2025**
- Built a data pipeline integrating structured (financial metrics) and unstructured (SEC filings) sources, enabling natural-language queries and automated insights for financial analysis and reporting.
- Implemented vector-based retrieval over financial documents to return context-grounded answers with source traceability.
- Prototyped AWS deployment using Bedrock, S3, and EC2, designing a scalable architecture for document retrieval, querying, and insight generation.

Financial Data Pipeline from SEC EDGAR and Inline XBRL (Arelle) **Nov 2024 - Jan 2025**
- Ingested SEC 10K/10Q filings via JSON and Inline XBRL, cleaned and normalized thousands of financial facts into structured wide tables, enabling accurate KPI tracking and downstream financial analysis.
- Implemented taxonomy fallbacks, unit normalization, and dimensional parsing to improve metric recovery, including EPS, and maintain consistent fiscal-year alignment across companies.
- Validated 1,100+ Inline XBRL facts from Apple 10-K filings against SEC JSON sources, producing clean CSV datasets for BI and financial analysis.

Realistic Rug Placement in Room Images (SAM-Guided Classical CV) **Jun 2025 - Jul 2025**
- Developed a data processing pipeline to manipulate and visualize image data with occlusion handling and perspective correction, showcasing ability to design end-to-end solutions that transform raw inputs into clear, stakeholder-ready outputs.
- Applied advanced image segmentation and transformation techniques (homography, shading, color transfer) to deliver realistic visual outputs, demonstrating skills in complex data processing and stakeholder-focused visualization.

EXPERIENCE

FuMind.ai **Mar 2026 - May 2026**
AI Engineering Intern *Remote Germany Startup*
- Contributed to the development and integration of AI models into production-oriented software workflows, supporting automation use cases across orchestrated services and backend systems.
- Supported workflow automation using orchestration tools to manage services, containers, and microservice-based components.
- Collaborated with engineering teams to improve deployment workflows, system reliability, and integration efficiency.

Tritone Analytics **Oct 2025 - Dec 2025**
Data Analyst Intern *Remote US Startup*
- Developed LLM and vision-LLM pipelines for extracting and normalizing structured financial data from PDFs and Excel files, ensuring numeric accuracy, null preservation, and schema consistency.

- Implemented data normalization and validation logic to standardize heterogeneous royalty and deduction statements into deterministic, machine-readable formats.
- Integrated financial domain knowledge and conversational memory into LLM workflows to accurately interpret, classify, and transform complex financial line items across documents.

AthleteRise **Oct 2025 - May 2026**
Fullstack AI Engineering *Remote India Based Startup*
- Contributed as an AI/ML and full-stack intern at AthleteRise, building computer-vision-based solutions for sports performance analysis.
- Worked on video-based cricket shot analysis, including preprocessing, pose/keypoint extraction, movement analysis, and model-assisted classification.
- Developed backend and frontend workflows for handling video inputs, running ML/CV inference, and presenting structured performance insights to users.

Samsung PRISM **Aug 2024 - Feb 2025**
Student Intern (Generative AI Image Enhancement)
- Analyzed generative neural networks (GNN) with a focus on model architecture and stochastic processes in image generation, enhancing understanding of data-driven AI solutions and anomaly detection insights.
- Implemented diffusion models to generate high-quality images through noise reduction and iterative refinement, demonstrating proficiency in machine learning algorithm testing and data transformation.
- Applied the Segment Anything Model (SAM) for image segmentation, optimizing object identification in complex images and contributing to enhanced data visualization practices.

SKILLS

- **Data & Analytics**: SQL, ETL Pipelines, Data Cleaning, Normalization, CSV/Parquet, Data Quality Checks, Excel, KPI Design, Dashboarding (Power BI, QuickSight, Tableau), Business Metrics & Trends, A/B Testing
- **Programming & Tools**: Python, Pandas, NumPy, Scikit-Learn, Data Visualization (Matplotlib, Seaborn), Power BI, Git, Docker, Java
- **Cloud & Platforms**: AWS (S3, EC2, Redshift), SageMaker, REST API
- **Statistical & Forecasting Methods**: Logistic Regression, Gradient Boosting, Time-Series Analysis, Calibration-, PyTorch, TensorFlow
- **Soft Skills**: Stakeholder communication, clear documentation, model cards, knowledge-sharing, disciplined delivery

CERTIFICATIONS

- Oracle Cloud Infrastructure 2024 Certified AI Foundations Associate: Oracle (2025)
- Introduction to Machine Learning: NPTEL, ELITE (2024)
- Responsible and Safe AI: NPTEL, ELITE (2024)
- Introduction to Database Systems: NPTEL

ACCOMPLISHMENTS

- 1st Rank in the Data Analytics Genesis Hackathon organized by IMI Genesis institute.
- 2nd rank in Data Odyssey Hackathon, TAPMI by analysing a dataseta and making rigorous financial predictions and stratergies.
- 3rd Rank in SRM's Trikon Hackathon by creating an AI-ecosystem enabled disaster relief and community engagement website.
- 2nd Rank in Data Samurai Data Science Competetion organized by SRM CINTEL club by structuring an improperly formatted tabular dataset and then Image reconstruction on noisy Images using Non Local Denoising Techiniques.
- Amongst the top 10 teams out of 2200 teams to be qualified for the second round of IIM Bangalore's Ideathon Flagship Event in 2024.

PUBLICATION

- Patent.*Published patent on an AI-driven augmented reality system for real-time navigation and object detection for visually impaired users. Publishing date : 21 Feb 2025, Application Number : 202541014793*
- Deep Learning Models for Classification of Red Blood Cells in Microscopy Images to Aid in Sickle Cell Anemia Diagnosis.-*Presented in 2025 1st International Conference on Smart and Intelligent Systems (SISCON) and Published in IEEE Xplore*

LANGUAGES

- English (Native or Professional Fluency)
- Hindi (Native or Professional Fluency)
- Japanese (Elementary Proficiency)